Predicting Residential Property Values Using Supervised Machine Learning for Real Estate Pricing Optimization

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Overview

Objective:

To develop a predictive model for estimating the tax-assessed value of homes using their physical and locational features, enhancing pricing strategies and buyer advising.

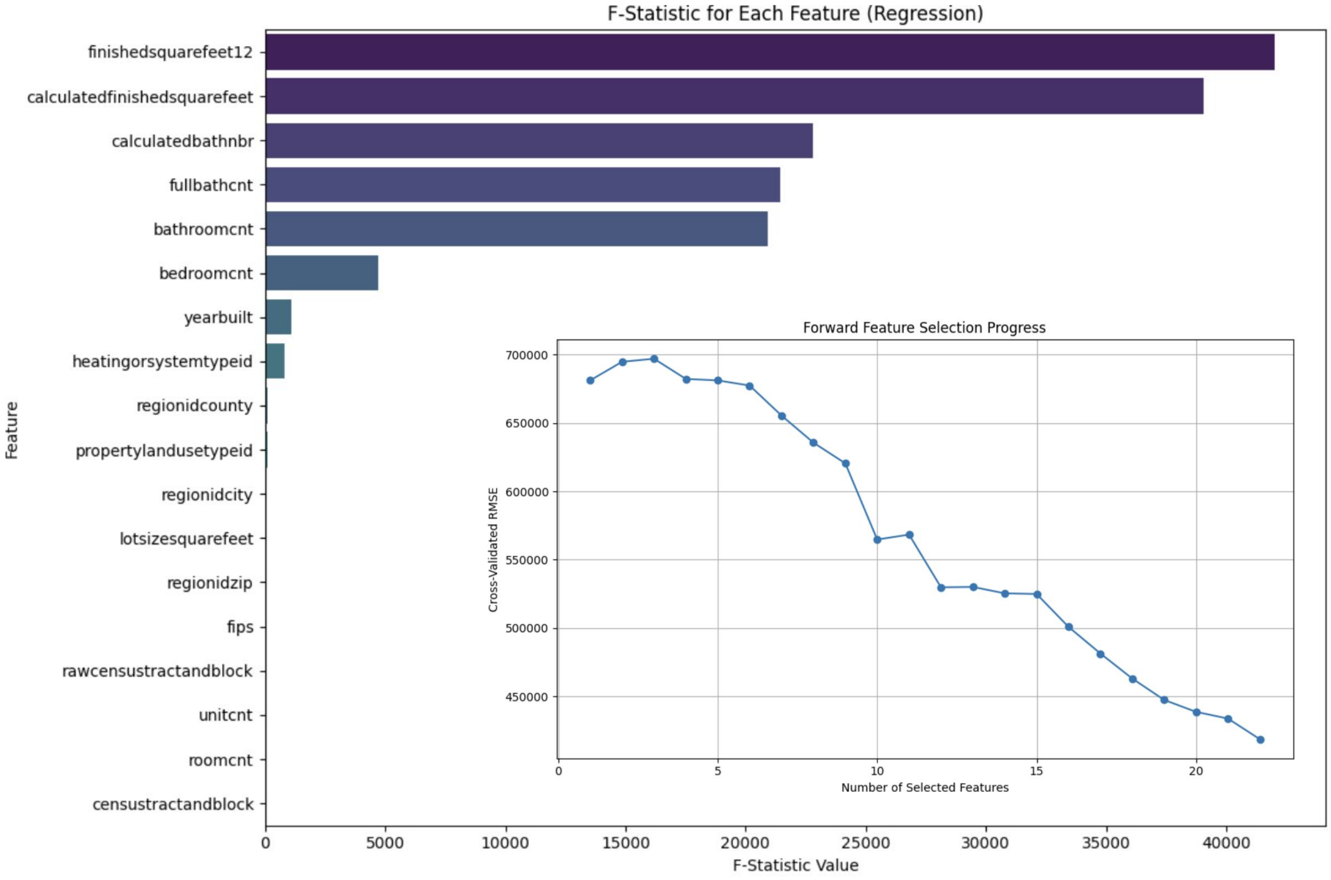
- **Method:** Regression models with RMSE optimization through extensive preprocessing and parameter tuning
- **Best Model:** Bagging Regressor (Test RMSE ≈ $417K)
- **Key Predictors:** Square footage, bathroom count
- **Next Steps:** Add external data (schools, crime rates); explore geospatial models

Zillow Housing Dataset

- 77,613 homes and 55 features
- Features with only one unique value unlikely to be useful
    - e.g., pool type, hot tub, deck type
- Location features had many unique values but high collinearity risk
- Features with few non-null values dropped due to sparsity
    - e.g., basementsqft, architecturalstyletypeid
- Latitude/longitude deemed unsuitable without geospatial modeling

Zillow Million Dollar Prize dataset (Kaggle, 2017)

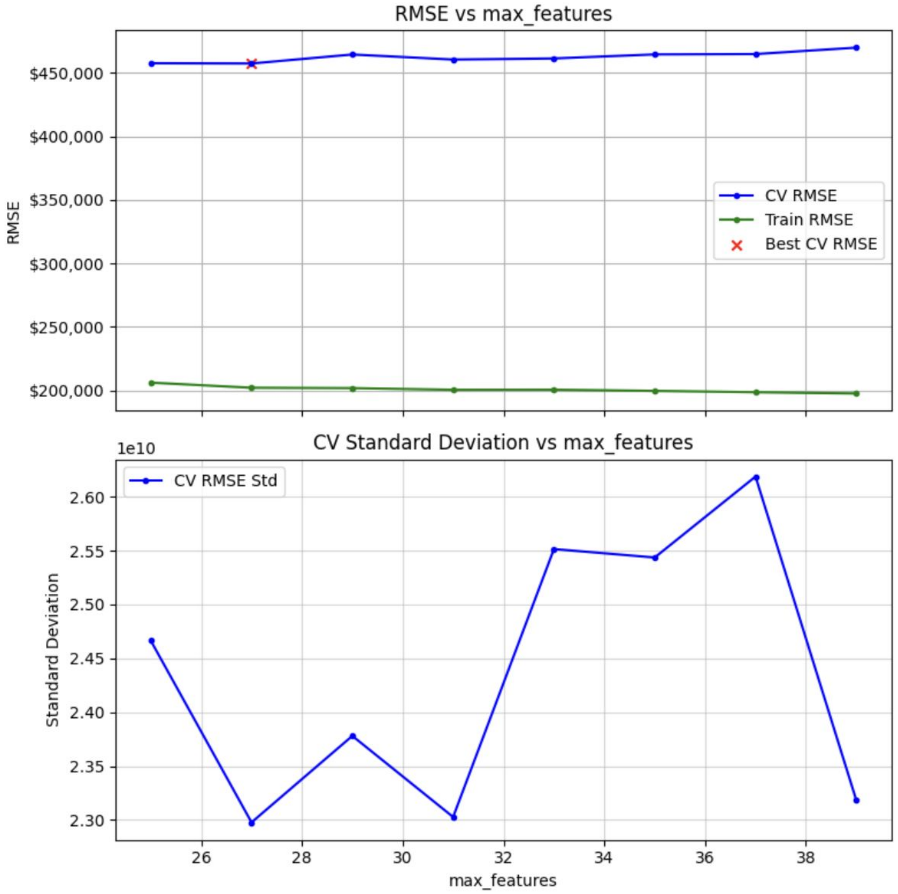
Methods: Data preparation



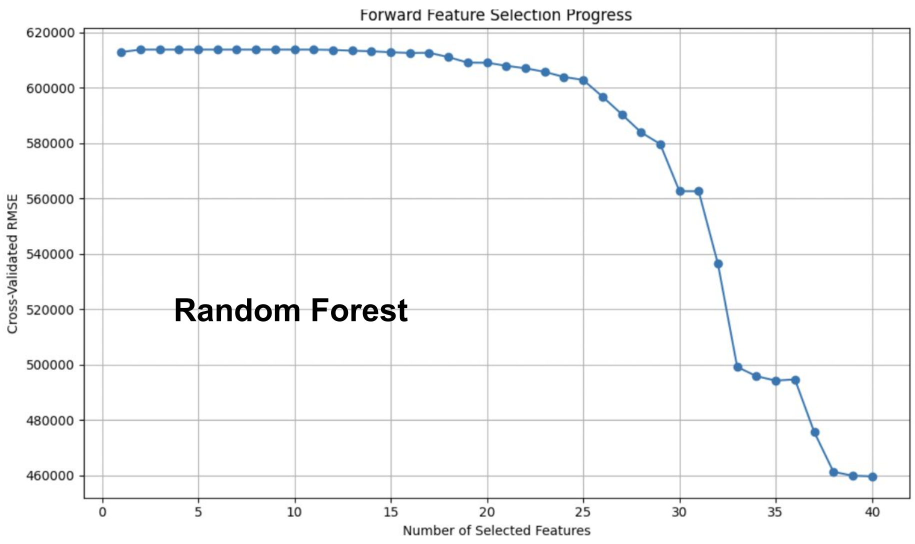
- **Duplicate IDs (n = 199)** identified and removed
- **Dropped features** with >40% missing values and those unsuitable for regression
- **Imputation**: Median for numeric, mode for categorical features
- **Feature selection**:
  - F-statistics (using f_regression)
  - Compared to forward and backward selection

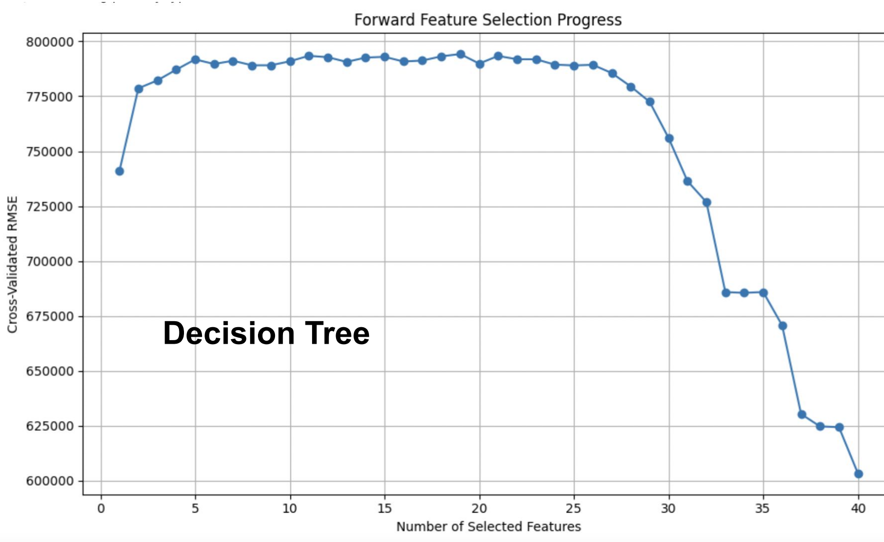
Methods: Model Selection

- **Hyperparameter tuning:**
  - Used sweep_parameter to narrow ranges
  - Validation curves plotted RMSE and CV Standard Deviation
- **Baseline modeling:**
  - 7 models tested using RepeatedKFold (5 splits, 5 repeats)
- **Feature engineering:**
  - Log transformation of “YearBuilt” to reduce skewness
  - Interaction term: ‘bedroomcnt’ × ‘bathroomcnt’
  - Polynomial feature: “calculatedfinishedsquarefeet”^{2}
- **Top Models:**
  - Bagging Regressor: Best RMSE, most interpretable
  - Random Forest: Second best in RMSE and stability
  - Gradient Boosting: Most complex, slightly higher RMSE



Results: Feature Selection - Forward Feature Selection





- **Best model:**
  - Random Forest Regressor
  - Mean RMSE: $447,186.13
  - STD RMSE: $19,661.73

- **Worst model:**
  - Decision Tree Regressor
  - Mean RMSE: $628,049.86
  - STD RMSE: $24,051.10

Results: Baseline Models

- Ensemble Tree based models performed better and more consistently than the regression models
- **Best Baseline Model:**
    - Random Forest Regressor (Mean RMSE: $445518.13)
- **Worst Baseline Model:**
    - Decision Tree Regressor (Mean RMSE: $608789.18)

Model	Baseline Mean RMSE ($)	Baseline Std RMSE ($)
Linear Regression	476342.01	23842.12
Ridge Regression	476316.76	23825.86
Lasso Regression	476340.63	23841.21
Decision Tree	608789.18	26988.88
Bagging Regressor	462458.49	20578.82
Random Forest	445518.13	20297.68
Gradient Boosting	450327.77	22309.71

Final Results for Optimized Bagging Regressor

- **GridSearchCV yielded final parameters:**
  - n_estimators = 485
  - max_samples = 0.3
  - max_features = 18
  - bootstrap = True
- Final GridSearch runtime: 7:02:45

	Train	Test
Mean CV RMSE	$396,113.36	$555,949.66
Standard Deviation CV RMSE	$132,117.42	$412,849.43
Best RMSE	$291,762.28	$417,517.70

Conclusions

Objective met: Successfully predicted home tax-assessed values using property features

- **Best model**: Bagging Regressor
  - Test RMSE ≈ $555,949
- **Key predictors**: Square footage, bathroom count
- **Business impact**:
  - Enables faster, more accurate pricing
  - Supports targeted marketing and tailored client advising

- **Limitations**:
  - High computational cost of hyperparameter tuning
  - Sparse features and missing data challenges
  - Non-quantifiable factors (e.g., neighborhood desirability) not fully captured

***Future work** will integrate external datasets (crime rates, school quality), improve feature selection strategies, and explore geospatial modeling for better location analysis*